



SalinBloc™

Containerised Water Treatment Platform

SalinBloc™ is Hydrovolta's standardised modular deployment platform for brackish groundwater treatment, pilot validation, and distributed water supply. Built around the SonixED™ electrodialysis reversal + ultrasound antifouling core, each unit arrives pre-commissioned with full cloud telemetry and autonomous operation from day one. No specialist on-site operator required.

STANDARD CONFIGURATIONS

Parameter	20 ft SalinBloc™	40 ft SalinBloc™
Flow capacity	15–45 m³/h	40–120 m³/h
SonixED™ stacks	1 stack	2–3 stacks
TDS feed range	Up to 5,000 ppm	Up to 20,000 ppm
Water recovery	Up to 98%*	Up to 98%*
Energy consumption	0.2–0.8 kWh/m³	0.2–0.8 kWh/m³
Operating pressure	0.5–2.5 bar	0.5–2.5 bar
Temperature range	5–55 °C	5–55 °C
Scale pathway	Parallel unit docking	Parallel unit docking
Operation	Autonomous + cloud telemetry	Autonomous + cloud telemetry
Footprint	6.1 m × 2.4 m	12.2 m × 2.4 m

* Up to 98% water recovery in suitable applications, subject to source chemistry and operating parameters.

PRIMARY APPLICATIONS

Municipal Utilities

Nitrate & salinity compliance

Large-Scale Agriculture

Soil-safe water for irrigation

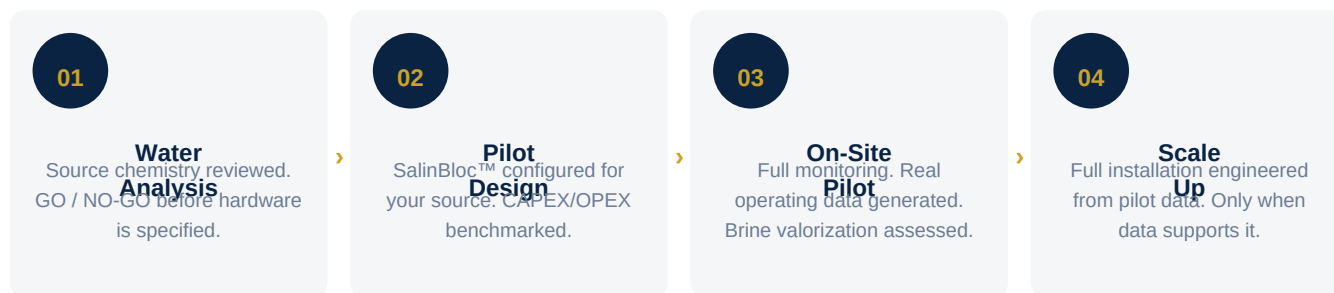
Remote Communities

Autonomous supply
no on-site operator

Industrial Process Water

Consistent quality
near-ZLD brine

DEPLOYMENT PATHWAY



INSIDE EVERY SALINBLOC™ UNIT

- **SonixED™ Stack**
Full EDR + ultrasound antifouling membrane stack. Patent 2021. Validated at Water Corporation Perth (3 utility sites).
- **Titanium Electrodes**
Corrosion-resistant electrode arrays for stable long-term operation in saline environments.
- **PLC Control System**
Automated polarity reversal, setpoint control, alarm handling. Minimal operator intervention required.
- **Cloud Telemetry**
Real-time data logging, remote monitoring, and performance dashboards from commissioning day one.
- **HMI Touch Panel**
7" operator interface for local monitoring, parameter adjustment.
- **Compact Pretreatment**
Integrated sediment filtration and pretreatment matched to source.

VALIDATED PERFORMANCE — GERMANY INDUSTRIAL PILOT 2023

90–98% recovery · 0.19–0.7 kWh/m³ · No anti-scalant · 43,680 m³/yr discharge reduction · \$0.30–0.40/m³ total cost

Compared against RO and standard EDR on same feed water over 90 days. Cooling tower blowdown, chemical industry.

Request SalinBloc™ specifications or pilot assessment

Send source water data (TDS, hardness, nitrate, silica, target flow) to: george@hydrovolta.com hydrovolta.com/contact